Gross Profit

	Three Months Ended			Variation	
	December 31, 2019	September 29, 2018	December 31, 2018	Sequential	Year-Over-Year
	(Unaudited, in millions)				
Gross profit	$ 1,081	$ 967	$ 1,059	11.8%	2.0%
Gross margin (as percentage of net revenues)	39.3%	37.9%	40.0%	+140 bps	-70 bps

Fourth quarter gross profit was $1,081 million and gross margin was 39.3%. On a sequential basis, gross margin increased 140 basis points, mainly driven by improved product mix and better manufacturing efficiencies.

Gross margin decreased 70 basis points year-over-year, mainly impacted by price pressure and unsaturation charges, partially offset by improved manufacturing efficiencies, better product mix and favorable currency effects, net of hedging.

Operating expenses

	Three Months Ended			% Variation	
	December 31, 2019	September 29, 2019	December 31, 2018	Sequential	Year-Over-Year
	(Unaudited, in millions)				
Selling, general and administrative expenses	$ (285)	$ (267)	$ (285)	(6.3)%	0.4%
Research and development expenses	(387)	(362)	(345)	(7.0)	(12.3)
Total operating expenses	**$ (672)**	**$ (629)**	**$ (630)**	**(6.7)%**	**(6.6)%**
As percentage of net revenues	(24.4)%	(24.7)%	(23.8)%	+30 bps	-60 bps

The amount of our operating expenses increased by $43 million on a sequential basis, mainly driven by seasonality and salary dynamic.

On a year-over-year basis, our operating expenses increased by $42 million, mainly due to salary dynamic and increased spending on certain R&D programs, partially offset by favorable currency effects, net of hedging.

Fourth quarter 2019 R&D expenses were net of research tax credits in France and Italy, which amounted to $37 million, compared to $29 million in the third quarter of 2019 and $39 million in the fourth quarter of 2018.

Other income and expenses, net

	Three Months Ended		
	December 31, 2019	September 29, 2019	December 31, 2018
	(Unaudited, in millions)		
Research and development funding	$ 68	$ 14	$ 19
Phase-out and start-up costs	(16)	(15)	(1)
Exchange gain (loss), net	1	(1)	—
Patent costs	(1)	(1)	(1)
Gain on sale of non-current assets	1	—	1
Other, net	1	1	(2)
Other income and expenses, net	**$ 54**	**$ (2)**	**$ 16**
As percentage of net revenues	2.0%	(0.1)%	0.6%